Transshipment and Assignment

A multinational company procures goods from 4 global production centres and distributes them to 6 different national distribution centres through 3 intermediate warehouses. The transportation costs between production centres, warehouses, and distribution centres are shown in the tables below

(a)

	Warehouse 1	**Warehouse 2**	**Warehouse 3**
Production Centre A	4	3	5
Production Centre B	6	2	4
Production Centre C	5	4	3
Production Centre D	7	5	6

Table 4.1 Transportation costs from production centres to warehouses (£ per unit)

	Distribution Centre					
	DC1	**DC 2**	**DC 3**	**DC 4**	**DC 5**	**DC 6**
Warehouse 1	3	2	5	4	6	7
Warehouse 2	1	4	2	3	5	6
Warehouse 3	5	3	4	2	3	4

Table 4.2 Transportation costs from warehouses to distribution centres (£ per unit)

Production centre	Supply
A	70
B	90
C	80
D	60

Table: Supply from the production centre

Distribution centre	Demand
1	**50**
2	**40**
3	**60**
4	**70**
5	**50**
6	**30**

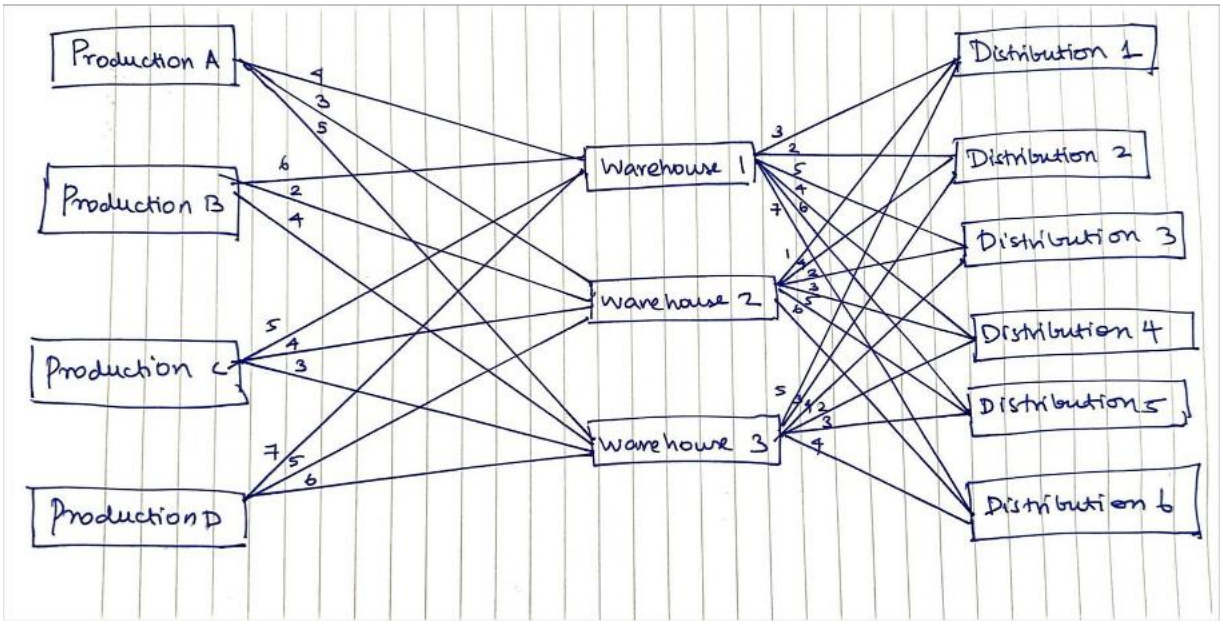
Table: Demand of the distribution centres

Problem statement:

Multinational company get goods from 4 production centres (A, B, C, D) and distributes to 6 different centres (DC1, DC2, DC3, DC4, DC5, DC6) through 3 warehouses (Warehouse 1, Warehouse 2, and Warehouse 3). the company wants to optimise its supply chain network by creating cost effective plan that minimises the total transportation cost. We use Linear programming (LP), specifically transportation optimisation method to minimise the transportation cost while satisfying the supply and demand.

Decision variable:

As shown in the diagram below, we can see there are 30 potential routes that can be used i.e. 30 decision variables.



x_{ij} = Quantity of goods transported from production centre i to warehouse j.

x_{jk} = Quantity of goods transported from warehouse j to distribution centre k

where, $i = \{A, B, C, D\}$; $j = \{1, 2, 3\}$

$k = \{1, 2, 3, 4, 5, 6\}$

Objective function

Minimize Cost $Z = \sum_{i,j} c_{ij} \cdot x_{ij} + \sum_{j,k} d_{jk} \cdot x_{jk}$

where, c_{ij} = cost of transporting one unit from i to j

d_{jk} = cost of transporting one unit from j to k

Supply and demand constraints

We create constraints to ensure the supply from production centre satisfies the demand required at distribution centres. We also found that the demand and the supply are 300, making this a balanced model.

flow constraint :-

$$\sum_{i} x_{ij} = \sum_{k} x_{jk} \quad \text{for all } j = \{1, 2, 3\}$$

non-negativity constraints $x_{ij} \geq 0$, $x_{jk} \geq 0$

Total mathematical representation:

```
Minimize Z = 4xa1 + 3xa2 + 5xa3 + 6xb1 + 2xb2 + 4xb3 + 5xc1 + 4xc2 +
             3xc3 + 7xd1 + 5xb2 + 6xd3 + 3x1d1 + 2x1d2 + 5x1d3 + 4x1d4 + 6x1d5 +
             7x1d6 + 1x2d1 + 4x2d2 + 2x2d3 + 3x2d4 + 5x2d5 + 6x2d6 + 5x3d1 + 3x3d2 +
             4x3d3 + 2x3d4 + 3x3d5 + 4x3d6
Subject to:

supply constraint

(xa1 + xa2 + xa3  == 70), "Supply_Constraint_A"
(xb1 + xb2 + xb3 == 90), "Supply_Constraint_B"
(xc1 + xc2 + xc3 == 80), "Supply_Constraint_C"
(xd1 + xd2 + xd3 == 60), "Supply_Constraint_D"

demand constraint

(x1d1 + x2d1 + x3d1 == 50), "Demand_Constraint_distribution_1"
(x1d2 + x2d2 + x3d2 == 40), "Demand_Constraint_distribution_2"
(x1d3 + x2d3 + x3d3 == 60), "Demand_Constraint_distribution_3"
(x1d4 + x2d4 + x3d4 == 70), "Demand_Constraint_distribution_4"
(x1d5 + x2d5 + x3d5 == 50), "Demand_Constraint_distribution_5"
(x1d6 + x2d6 + x3d6 == 30), "Demand_Constraint_distribution_6"

Flow constraint

(xa1 + xb1 + xc1 + xd1 - x1d1 - x1d2 - x1d3 -x1d4 - x1d5 - x1d6 == 0), "Flow_Balance_warehouse_1"
(xa2 + xb2 + xc2 + xd2 - x2d1 - x2d2 - x2d3 -x2d4 - x2d5 - x2d6 == 0), "Flow_Balance_warehouse_2"
(xa3 + xb3 + xc3 + xd3 - x3d1 - x3d2 - x3d3 -x3d4 - x3d5 - x3d6 == 0), "Flow_Balance_warehouse_3"
```

(a) Build a model and solve it to help the multinational company optimize the supply chain network

Solution:

```
x1d1 = 0.0
x1d2 = 40.0
x1d3 = 0.0
x1d4 = 0.0
x1d5 = 0.0
x1d6 = 0.0
x2d1 = 50.0
x2d2 = 0.0
x2d3 = 60.0
x2d4 = 10.0
x2d5 = 0.0
x2d6 = 0.0
x3d1 = 0.0
x3d2 = 0.0
x3d3 = 0.0
x3d4 = 60.0
x3d5 = 50.0
x3d6 = 30.0
xa1 = 40.0
xa2 = 30.0
xa3 = 0.0
xb1 = 0.0
xb2 = 90.0
xb3 = 0.0
xc1 = 0.0
xc2 = 0.0
xc3 = 80.0
xd1 = 0.0
xd2 = 0.0
xd3 = 60.0
Value of Objective Function = 1700.0
```

(b)) **Assume the operating costs of Warehouse 2 have increased, and the cost of transporting goods through this warehouse rises by £1 per unit. Rebuild the model and solve it under the new conditions.**

```
x1d1 = 0.0
x1d2 = 40.0
x1d3 = 0.0
x1d4 = 0.0
x1d5 = 0.0
x1d6 = 0.0
x2d1 = 50.0
x2d2 = 0.0
x2d3 = 60.0
x2d4 = 0.0
x2d5 = 0.0
x2d6 = 0.0
x3d1 = 0.0
x3d2 = 0.0
x3d3 = 0.0
x3d4 = 70.0
x3d5 = 50.0
x3d6 = 30.0
xa1 = 40.0
xa2 = 30.0
xa3 = 0.0
xb1 = 0.0
xb2 = 80.0
xb3 = 10.0
xc1 = 0.0
xc2 = 0.0
xc3 = 80.0
xd1 = 0.0
xd2 = 0.0
xd3 = 60.0
Value of Objective Function = 1930.0
```

(c) Due to changes in trade policies, the company wants the handling capacity of Warehouse 1 not to exceed 50 units. Rebuild the model and solve it.

```
x1d1 = 0.0
x1d2 = 40.0
x1d3 = 0.0
x1d4 = 0.0
x1d5 = 0.0
x1d6 = 0.0
x2d1 = 50.0
x2d2 = 0.0
x2d3 = 60.0
x2d4 = 10.0
x2d5 = 0.0
x2d6 = 0.0
x3d1 = 0.0
x3d2 = 0.0
x3d3 = 0.0
x3d4 = 60.0
x3d5 = 50.0
x3d6 = 30.0
xa1 = 40.0
xa2 = 30.0
xa3 = 0.0
xb1 = 0.0
xb2 = 90.0
xb3 = 0.0
xc1 = 0.0
xc2 = 0.0
xc3 = 80.0
xd1 = 0.0
xd2 = 0.0
xd3 = 60.0
Value of Objective Function = 1700.0
```

(d)The company wants to remove one distribution center (DC1 or DC5) and then equally assign the original demand to others. Rebuild the model and find the best plan.

If the company removes Distribution centre 1and redistributes the demand equally

```
x1d1 = 0.0
x1d2 = 50.0
x1d3 = 0.0
x1d4 = 0.0
x1d5 = 0.0
x1d6 = 0.0
x2d1 = 0.0
x2d2 = 0.0
x2d3 = 70.0
x2d4 = 80.0
x2d5 = 0.0
x2d6 = 0.0
x3d1 = 0.0
x3d2 = 0.0
x3d3 = 0.0
x3d4 = 0.0
x3d5 = 60.0
x3d6 = 40.0
xa1 = 50.0
xa2 = 20.0
xa3 = 0.0
xb1 = 0.0
xb2 = 90.0
xb3 = 0.0
xc1 = 0.0
xc2 = 0.0
xc3 = 80.0
xd1 = 0.0
xd2 = 40.0
xd3 = 20.0
```

The minimized transportation cost is **£1820**

If the company removes Distribution centre 5and redistributes the demand equally

```
x1d1 = 0.0
x1d2 = 50.0
x1d3 = 0.0
x1d4 = 0.0
x1d5 = 0.0
x1d6 = 0.0
x2d1 = 60.0
x2d2 = 0.0
x2d3 = 70.0
x2d4 = 0.0
x2d5 = 0.0
x2d6 = 0.0
x3d1 = 0.0
x3d2 = 0.0
x3d3 = 0.0
x3d4 = 80.0
x3d5 = 0.0
x3d6 = 40.0
xa1 = 50.0
xa2 = 20.0
xa3 = 0.0
xb1 = 0.0
xb2 = 90.0
xb3 = 0.0
xc1 = 0.0
xc2 = 0.0
xc3 = 80.0
xd1 = 0.0
xd2 = 20.0
xd3 = 40.0
```

The minimized transportation cost is **£1640**

Insights and recommendations:

Several significant findings are also derived when the distribution network optimization analysis is considered. Dropping Distribution Centre 5 to spread its demand over the rest of the centres yields the least expensive price cost of £1640, suggesting that Centre 5 was a less efficient resource or a more expensive operation than the remainder. This is consistent with the conclusion that the cost-optimum choice removes Centre 5. For Warehouse 1, limited to 50 units, the overall cost is constant at £1700, which suggests that restricting its capacity can avoid inefficiency and, hence, distribute the goods without significant cost increases. This clearly demonstrates the need to control warehouse capacities to obtain minimum cost.

Nevertheless, if the operating cost is increased by a 1 at Warehouse 2, it will increase the total cost to £1930, which is thus a less favourable choice. This shows that Warehouse 2 is sensitive to changes in the operating cost, indicating that some cost control measures should be implemented to keep the total efficiency high. Also removing Distribution Centre 1 increases the base cost to £1820, showing that although it is not the optimal model, it offers cost reductions. Therefore, Centre 5 remains the best option for elimination.